

Abhinav Rao

Computer Science Ph.D. Student, University of Maryland, College Park, MD

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Education

University of Maryland at College Park

College Park, MD

Ph.D. student, Computer Science (Area: AI-Security, Advisor: Yizheng Chen)

August 2025 - Present

Carnegie Mellon University | Language Technologies Institute

Pittsburgh, PA

M.S. student in NLP (MHS, Research: Responsible-AI, AI safety)

August 2023 - December 2024

Birla Institute of Technology and Science (BITS) Pilani

Hyderabad, India

B.E. Computer Science (Graduated Early)

August 2018 - February 2022

Select Experience

Language Technologies Institute

Pittsburgh, PA

Systems Software Engineer (Technical report)

Jan 2025 - July 2025

▷ Developed the technical approach for winning proposal, securing \$250,000 for the [Amazon NOVA AI Challenge](#). Employed [deliberative-alignment](#) and reinforcement learning (GRPO) to enable secure code reasoning. Achieved defense success of ~90% in the final red-teaming tournament.

Microsoft

Bangalore, India

Research Fellow

Jan 2022 - August 2023

▷ Responsible AI (RAI) research focusing on AI Ethics and Safety. Published on (a) ethical reasoning capabilities of LLMs, (b) and LLMs' susceptibility to jailbreaks.

Select Publications

[1.] *Tricking LLMs into Disobedience: Formalizing, Analyzing, and Detecting Jailbreaks* (**Abhinav Rao**, Sachin Vashistha*, Atharva Naik*, Somak Aditya, and Monojit Choudhury) [LREC-CoLING '24]

[2.] *Ethical Reasoning over Moral Alignment: A Case and Framework for In-Context Ethical Policies in LLMs* (**Abhinav Rao***, Aditi Khandelwal*, Kumar Tanmay*, Utkarsh Agarwal*, Monojit Choudhury) [Findings of EMNLP '23, Keynote @WiNLP '23]

[3.] *NormAd: A framework for measuring the cultural adaptability of large language models* (**Abhinav Rao***, Akhila Yerukola*, Vishwa Shah, Katharina Reinecke, Maarten Sap) [NAACL '25, C3NLP@ACL'24]

Ongoing Research Projects

Secure and Correct Code Generation

September 2025 - Present

Advisor: Prof. Yizheng Chen

– Building a 30K-prompt, multi-language reinforcement learning dataset spanning ~25 CWEs and training Qwen3-8B with process rewards to improve secure code generation and reasoning on three benchmark suites.

Skills

AI & Security: LLM and Agent Security, AI Safety, LLM Red Teaming, Program Analysis, Vulnerability Detection.

Machine Learning: Reinforcement Learning (RLHF, RLAI, GRPO), PyTorch, Hugging Face, Transformers, vLLM, Python, Slurm